MATH PAPER 2

Compulsory Part · Multiple Choice

Mock Examination — Paper 2

Paper 2 (MC)

75 minutes (1.25 hours)

FOR REVISION

For revision and self-assessment only. This is not an official examination paper.

Difficulty rating：**3.5** / 5.0

INSTRUCTIONS

1. This section contains multiple-choice questions; each has ONE correct answer.
2. All questions carry equal marks.
3. Choose only ONE answer for each question. More than one answer scores NO MARKS.
4. No marks will be deducted for wrong answers.

Independent, unofficial mock paper aligned to the DSE syllabus. Not affiliated with any official examination authority. For revision only.

There are 30 questions in Section A and 15 questions in Section B.
The diagrams in this paper are not necessarily drawn to scale. Choose the best answer for each question.

Section A

1. Expand and simplify $(2x+3)^2+(x-4)^2$, giving the answer in descending powers of x.

A. $5x^2+4x+25$

B. $5x^2-4x+25$

C. $5x^2+4x+7$

D. $3x^2+4x+25$

2. The length of a rope is measured as 48 cm and the relative error is $\frac{1}{96}$. Find the maximum absolute error.

A. 0.05 cm

B. 0.25 cm

C. 0.4 cm

D. 0.5 cm

3. In the rectangular coordinate plane, the point $P(7,-2)$ is rotated through $180°$ about the point $(3,1)$ to P'. Find the coordinates of P'.

A. $(1,-4)$

B. $(-1,4)$

C. $(-7,2)$

D. $(4,-1)$

4. It is given that y varies inversely as x. When $x=6$, $y=15$. Find the value of y when $x=10$.

A. 4

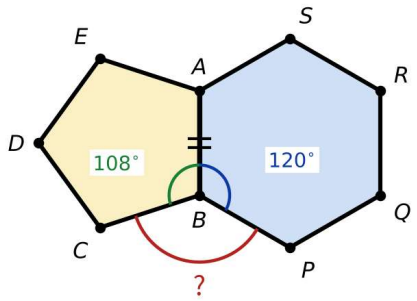
B. 6

C. 8

D. 9

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5. In the figure, the regular pentagon $ABCDE$ and the regular hexagon $ABPQRS$ have a common side AB and lie on opposite sides of AB. Find $\angle CBP$.



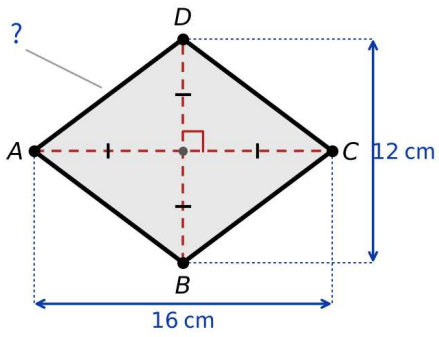
A. $120°$

B. $132°$

C. $138°$

D. $150°$

6. In the figure, $ABCD$ is a rhombus-shaped tile whose diagonals are of lengths 16 cm and 12 cm. Find the perimeter of the tile.



A. 36 cm

B. 40 cm

C. 48 cm

D. 56 cm

7. Two numbers x and y satisfy both $2x + 3y = 12$ and $x - y = 1$. If x and y also satisfy $kx + y = 11$, find the value of the constant k.

A. 3

B. 4

C. 5

D. 9

8. Without finding the value of n, simplify $\frac{8^{n+1}}{4^n}$, where n is a positive integer, giving the answer as a power of 2.

A. 2^{n+3}

B. 2^{n+1}

C. 2^{n+2}

D. 2^{3n+3}

9. Convert the decimal number 205 into a binary number.

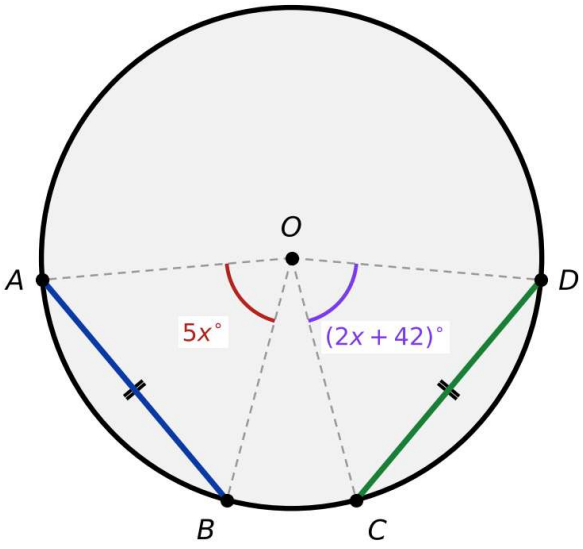
A. $(11001101)_2$

B. $(11001110)_2$

C. $(11010101)_2$

D. $(110011001)_2$

10. In the figure, O is the centre and AB and CD are two chords of the circle O, where $AB = CD$. It is given that $\angle AOB = 5x°$ and $\angle COD = (2x + 42)°$. Find the value of x, giving the answer as an integer.



A. 8

B. 12

C. 14

D. 18

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11. A sum of **24000** dollars is deposited at simple interest and the interest obtained after **3** years is **2880** dollars. Find the interest rate per annum.

A. **1%**

B. **2%**

C. **3%**

D. **4%**

12. It is given that α and β are the two real roots of the equation $x^2 - 7x + 10 = 0$. Without solving the equation, find the value of $(\alpha - \beta)^2$.

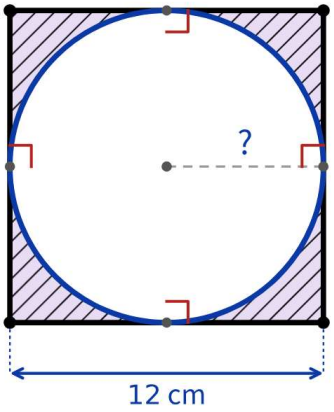
A. -9

B. 0

C. 4

D. 9

13. In the figure, the shaded region is formed by removing its inscribed circle from a square of side **12** cm. Find the area of the shaded region in terms of π.



A. $(36 - 9\pi)$ cm^2

B. $(72 - 18\pi)$ cm^2

C. $(108 - 27\pi)$ cm^2

D. $(144 - 36\pi)$ cm^2

14. Express $\frac{3}{x-2}+\frac{2}{x+3}$ as a single fraction and factorize its numerator completely, where x is neither 2 nor -3.

A. $\frac{5}{(x-2)(x+3)}$

B. $\frac{5(x+1)}{(x-2)(x+3)}$

C. $\frac{5(x-1)}{(x-2)(x+3)}$

D. $\frac{x+13}{(x-2)(x+3)}$

15. The median of the seven data 11, 4, 9, x, 6, 13 and 6 is 8, where x is an integer. Find the value of x.

A. 6

B. 7

C. 8

D. 9

16. The sum of the first 20 terms of an arithmetic series is 650 and the first term of the sequence is 5. Find the 20 th term of the sequence.

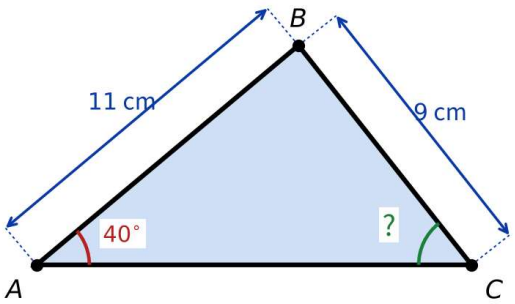
A. 35

B. 45

C. 60

D. 62

17. In the figure, $\triangle ABC$ is the plane figure of a triangular glass panel, where $AB = 11$ cm and $BC = 9$ cm. It is given that $\angle BAC = 40°$ and $\angle ACB$ is an acute angle. Without measuring any angle in the figure, find $\angle ACB$, correct to the nearest 0.1 degree.



A. $35.6°$

B. $40.0°$

C. $47.3°$

D. $51.8°$

貼條碼 / Barcode

18. A car consumes **7.5** L of fuel for every **100** km travelled. If there are **48** L of fuel in the tank, find the greatest distance that the car can travel.

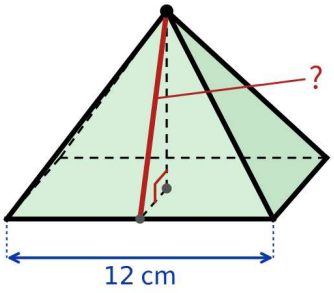
A. **640** km

B. **660** km

C. **720** km

D. **6400** km

19. The base of the solid right pyramid in the figure is a square of side **12** cm and the volume of the pyramid is **384** cm^3. Find the slant height of the pyramid in cm.



A. **8** cm

B. **9** cm

C. **10** cm

D. **12** cm

20. Without using a calculator, find the imaginary part of the complex number $(3-4i)^2$, where i is the imaginary unit and $i^2=-1$.

A. -48

B. -25

C. -24

D. -7

21. In the rectangular coordinate plane, the equation of the straight line L_1 is $2x-y+6=0$ and the equation of the straight line L_2 is $4x-2y-5=0$. Consider the following three statements:

I． L_1 and L_2 are parallel to each other.

II． The y-intercept of L_1 is 6.

III． L_2 passes through the point $(2,2)$.

Which of the above is/are correct?

A. 只有 III

B. 只有 I 及 II

C. 只有 I 及 III

D. 只有 II 及 III

22. A quadratic function is $y = -2x^2 + 12x - 5$, where x is any real number. Without drawing its graph, use the method of completing the square to find the maximum value of y.

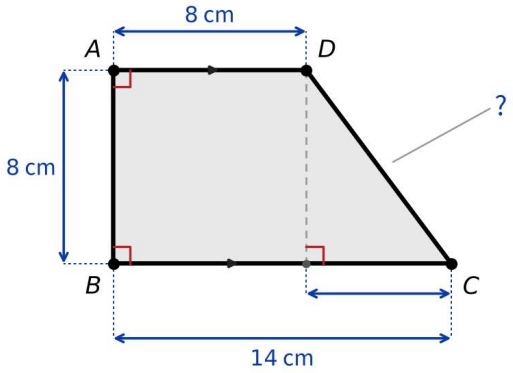
A. 3

B. 11

C. 13

D. 18

23. In the figure, $ABCD$ is a right-angled trapezium-shaped metal sheet with four straight sides, where AD is parallel to BC and $\angle ABC = \angle BAD = 90°$. It is given that $AD = 8$ cm, $BC = 14$ cm and $AB = 8$ cm. Find the perimeter of the metal sheet.



A. 36 cm

B. 38 cm

C. 40 cm

D. 42 cm

24. Let α and β be the roots of the equation $2x^2 - 5x - 3 = 0$. Without solving the equation, consider the following three statements:

I · The equation has two distinct real roots.

II · $\alpha + \beta = -\frac{5}{2}$

III · $\alpha\beta = -\frac{3}{2}$

Which of the above is/are correct?

A. 只有 III

B. 只有 I 及 II

C. 只有 I 及 III

D. 只有 II 及 III

貼條碼 / Barcode

25. A bag contains **8** red balls and some white balls. A ball is randomly drawn from the bag and the probability of drawing a red ball is $\frac{2}{5}$. Find the number of white balls.

A. **3**

B. **5**

C. **8**

D. **12**

26. The marked price of an article is set at **40%** above its cost and the selling price is **85%** of the marked price. It is given that the cost is **500** dollars. Consider the following three statements:

I · The marked price of the article is **680** dollars.

II · The selling price of the article is **595** dollars.

III · The profit percent of this sale is **25%** .

Which of the above is/are correct?

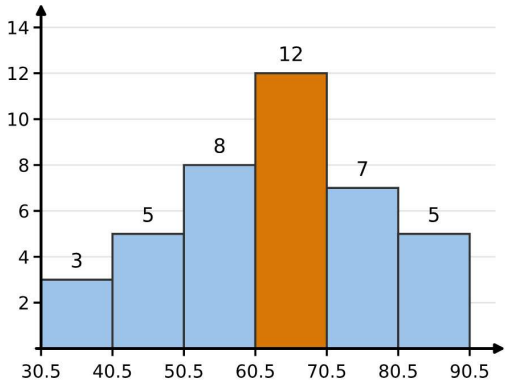
A. 只有 I

B. 只有 II

C. 只有 III

D. 只有 I 及 II

27. The frequency histogram in the figure shows the test scores of the students in a class, where the horizontal axis shows the score and the vertical axis shows the number of students. Find the percentage of students whose scores lie in the class from **60.5** marks to **70.5** marks.



A. **17.5%**

B. **20%**

C. **30%**

D. **33.3%**

28. In a right-angled triangle ABC, $\angle ABC = 90°$ and θ is $\angle BAC$. It is given that $\sin\theta = \frac{5}{13}$, where θ is an acute angle. Without first finding the value of θ, consider the following three statements:

I · $\cos\theta = \frac{8}{13}$

II · $\tan\theta = \frac{13}{5}$

III · $\sin\theta\cos\theta = \frac{60}{169}$

Which of the above is/are correct?

A. 只有 III

B. 只有 I 及 II

C. 只有 I 及 III

D. 只有 II 及 III

29. Let $f(x) = 2x^3 - 3x^2 - 11x + 6$, where x is any real number. Without using long division, use the remainder theorem and the factor theorem to consider the following three statements:

I · $f(3) = 0$

II · $x + 2$ is a factor of $f(x)$.

III · $f(x) = (x - 3)(x + 2)(2x - 1)$

Which of the above is/are correct?

A. 只有 I 及 II

B. 只有 I 及 III

C. 只有 II 及 III

D. I、II 及 III

30. The mean and the standard deviation of the test scores of a class are 66 marks and 8 marks respectively, and Ming scores 78 marks. Consider the following three statements:

I · The standard score of Ming is 1.2.

II · If 5 marks are added to the score of each student, the new standard deviation is still 8 marks.

III · If the score of each student is increased by 10%, the new standard deviation is 8.8 marks.

Which of the above is/are correct?

A. 只有 III

B. 只有 I 及 II

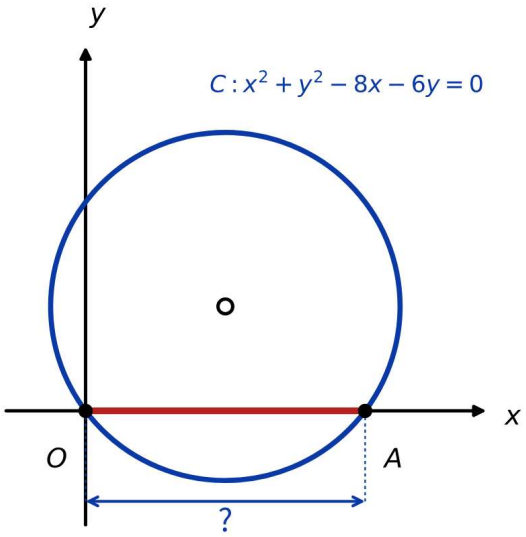
C. 只有 I 及 III

D. 只有 II 及 III

Section B

貼條碼 / Barcode

31. In the figure, in a rectangular coordinate plane with one unit length as unit, the equation of the circle C is $x^2+y^2-8x-6y=0$. It is given that C cuts the x-axis at the origin O and another point A, and the centre of C lies in quadrant I. Find the length of OA.



A. 4

B. 5

C. 6

D. 8

32. In the rectangular coordinate plane, a parabola cuts the x-axis at $(-1,0)$ and $(4,0)$, and its y-intercept is -12. Find the quadratic function represented by the parabola.

A. $y=x^2-3x-4$

B. $y=3x^2+9x-12$

C. $y=3x^2-9x-12$

D. $y=-3x^2+9x+12$

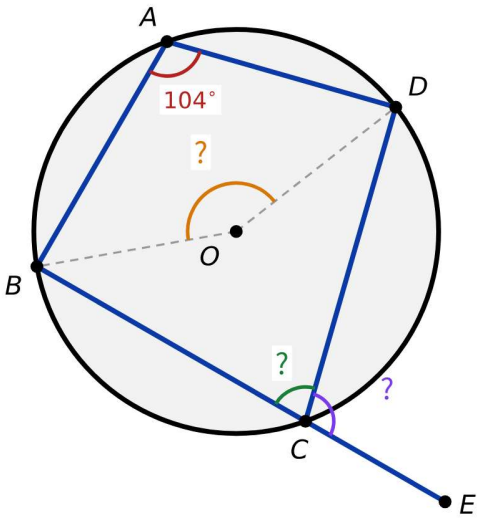
33. In the figure, A, B, C and D are four points on the circle with centre O and $ABCD$ is a cyclic quadrilateral. It is given that $\angle BAD = 104°$ and E is a point on BC produced such that E and B lie on opposite sides of C. Without measuring any angle in the figure (the figure is not drawn to scale), consider the following three statements:

I · $\angle BCD = 76°$

II · $\angle DCE = 76°$

III · $\angle BOD = 104°$, where $\angle BOD$ is not a reflex angle.

Which of the above is/are correct?



A. 只有 I

B. 只有 II

C. 只有 III

D. 只有 I 及 II

34. The common ratio of an infinite geometric series is $\frac{2}{5}$ and the sum of all the terms of the series is 45. Find the 3rd term of the geometric sequence corresponding to the series, giving the answer as a decimal.

A. **1.728**

B. **2.7**

C. **4.2**

D. **4.32**

貼條碼 / Barcode

35. In a polar coordinate system, O is the pole and the polar axis is along the positive direction of the x-axis. The polar coordinates of the point P are $(8, 150°)$. Without using a calculator, find the rectangular coordinates of P.

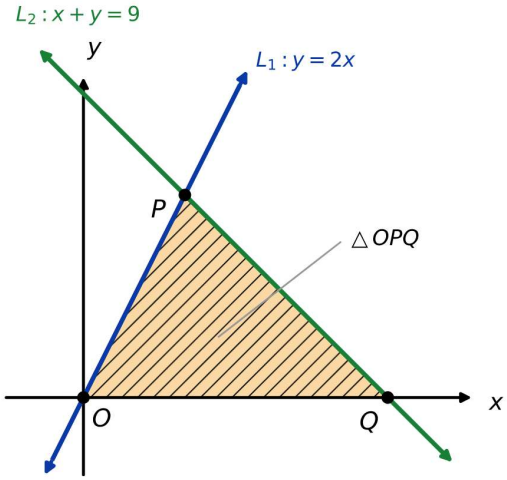
A. $(-4\sqrt{3}, 4)$

B. $(-4, 4\sqrt{3})$

C. $(4\sqrt{3}, -4)$

D. $(4, 4\sqrt{3})$

36. In the figure, the equation of the straight line L_1 is $y = 2x$ and the equation of the straight line L_2 is $x + y = 9$. L_1 and L_2 intersect at P, and L_2 cuts the x-axis at Q. Find the area of $\triangle OPQ$, where O is the origin, giving the answer in square units.



A. **24**

B. **27**

C. **31.5**

D. **40.5**

37. In a workshop, whether machine A breaks down and whether machine B breaks down are independent. On a certain day, the probability that machine A breaks down is **0.2** and the probability that machine B breaks down is **0.1**. Find the probability that at least one of the machines breaks down on that day.

A. **0.1**

B. **0.26**

C. **0.28**

D. **0.3**

38. A workshop produces chairs of type A and type B. Each type A chair requires **2** hours of sanding and **1** hour of painting, while each type B chair requires **1** hour of sanding and **2** hours of painting. The workshop provides at most **18** hours of sanding and **21** hours of painting each day. The profit on each type A chair is **120** dollars and the profit on each type B chair is **150** dollars. Find the greatest total profit that the workshop can make in a day.

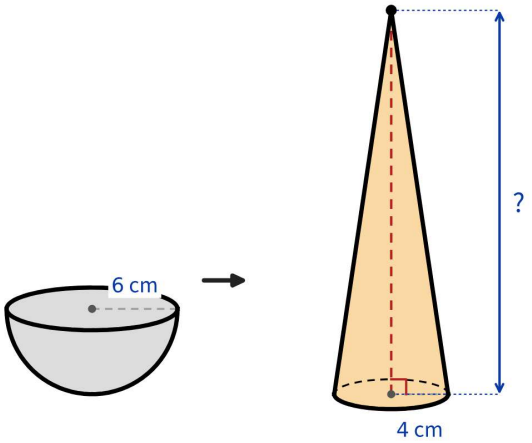
A. **1710** 元

B. **1800** 元

C. **1890** 元

D. **2160** 元

39. In the figure, a solid metal hemisphere of radius **6** cm is recast into a solid cone of base radius **4** cm, and no metal is lost in the process. Find the height of the cone.



A. **27** cm

B. **36** cm

C. **54** cm

D. **81** cm

40. The complex number z satisfies $(2+i)z = 5+10i$, where i is the imaginary unit and $i^2 = -1$. Without using a calculator, find z, giving the answer in the form $a+bi$, where a and b are real numbers.

A. $4+3i$

B. $4-3i$

C. $3+4i$

D. $-4+3i$

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41. Three positive integers are in arithmetic sequence. Their sum is 42 and their product is 2688 . Without testing the integers one by one, find the largest of the three numbers.

A. 16

B. 18

C. 21

D. 24

42. Consider the following three statements about loci on a plane:

I． The locus of a point at a distance of 5 cm from a fixed point A is a circle with centre A and radius 5 cm.

II． The locus of a point at a distance of 3 cm from a fixed straight line L is a straight line parallel to L .

III． The locus of a point inside a fixed angle and equidistant from the two sides of the angle is the angle bisector of that angle.

Which of the above is/are correct?

A. 只有 I 及 II

B. 只有 I 及 III

C. 只有 II 及 III

D. I、II 及 III

43. It is given that x is a real number and $4^{x+1} = 8^{x-1}$. Without using logarithms or a calculator, first express both sides as powers of 2 and then find the value of x .

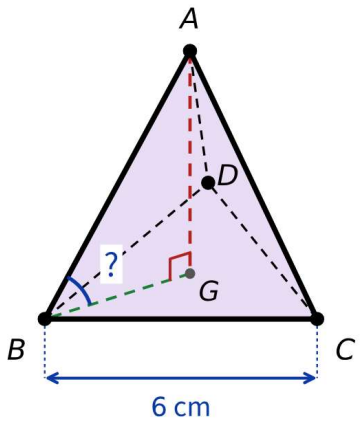
A. -5

B. 0

C. 3

D. 5

44. In the figure, $ABCD$ is a regular tetrahedron of edge 6 cm whose four faces are congruent equilateral triangles. Find the angle between the edge AB and the base BCD, correct to the nearest 0.1 degree.



A. $45.0°$

B. $54.7°$

C. $60.0°$

D. $70.5°$

45. It is given that $\log 2 = 0.301$ and $\log 3 = 0.477$, both correct to 3 decimal places, where all the logarithms are common logarithms with base 10. Without using a calculator and using only the two approximations above, consider the following three statements (the value of each logarithm below is correct to 3 decimal places):

I · $\log 6 = 0.778$

II · $\log 12 = 1.079$

III · $\log \frac{3}{2} = 1.585$

Which of the above is/are correct?

A. 只有 III

B. 只有 I 及 II

C. 只有 I 及 III

D. 只有 II 及 III

END OF PAPER